

PhD Reference Material

Personal Physics Notes

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Preface

These notes are a record of concepts, techniques, foundations, and connections that I found important to understand as I moved through my PhD.

Notation

Linear Algebra

$|\psi\rangle$ Ket vector.

$\langle\psi|$ Bra vector under the Hermitian inner product.

$|\psi)$ Ket vector when working with the c-product.

$(\psi|$ Bra vector when working with the c-product.

Part I

Mathematical Foundations

Chapter 1

Sets and Mathematical Preliminaries

1.1 General Terminology

- Set: a collection of mathematical objects. X .
- Null / empty set: The set defined by containing nothing. $\emptyset$.
- Map: a mapping that associates objects from one set X to objects of another Y . $m : X \rightarrow Y$.
- Product: $X \times Y$ is the set of all ordered pairs (x, y) where $x \in X$, $y \in Y$. That is, $X \times Y = \{(a, b) \mid x \in X \text{ and } y \in Y\}$.

1.2 Extra Set Terminology

- Subset: a set U that contains some or all of the elements of another set X : $U \subseteq X$.
- Proper subset: a set U that contains some, but not all, of the elements of another set X : $U \subset X$.
- Intersection: the set made of the items between two sets X and Y . $X \cap Y$.
- Union: the set made of combining two sets. $X \cup Y$.

1.3 Extra Function Terminology

- Domain: the set a mapping is restricted to act on. E.g. for $m : U \subset X \rightarrow Y$, $\text{Dom}(m) = U$.
- Codomain: the set a mapping is restricted to map into. E.g., for $m : X \rightarrow Y$, $\text{Cod}(m) = Y$.
- Image of a subset $U \subseteq X$ for a mapping $m : X \rightarrow Y$: $m(U) = \{m(u) \mid u \in U\}$. $\text{Image}(m) = m(X)$.
- Inverse: if $m : X \rightarrow Y$, then $m^{-1}(U) = \{x \mid x \in X \text{ and } m(x) \in U\}$ for $U \subseteq Y$.
- Function: a mapping that assigns exactly one element of the second set to each element of the first.
- Bijection: a mapping that is both injective and surjective.

- Injective (one-to-one): A function that maps distinct elements of its domain to distinct elements of its codomain: that is, $x_1 \neq x_2$ implies $m(x_1) \neq m(x_2)$.
- Surjective (onto): A function for which, for every element y of the function's codomain, there exists at least one element x in the function's domain such that $f(x) = y$.

1.4 Equivalence Classes and Quotient Spaces

- Binary relation over sets X, Y is a set of ordered pairs (x, y) , $x \in X$, $y \in Y$ (or, $(x, y) \in X \times Y$). Intuitively, it represents the concept of a relation: an element x is *related* to an element y if and only if the pair (x, y) belongs to the set of ordered pairs that defines the binary relation.
- Equivalence relation on a set X : a binary relation $\sim$ on X satisfying the following properties:
 - $a \sim a$ for all $a \in X$ (reflexivity),
 - $a \sim b$ implies $b \sim a$ for all $a, b \in X$ (symmetry),
 - $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in X$ (transitivity).
- Equivalence class of an element $a \in X$: the set $[a] = \{x \in X : a \sim x\}$.
- Quotient set of X by R : If X is a set and R is an equivalence relation, then the quotient set of X by R , X/R , is the set of all equivalence classes in X with respect to an equivalence relation R . This is often written as $X/\sim$ for an understood equivalence relation written as $\sim$.

1.5 Structure and Homomorphisms

We can add structure, or **operations**, to a set. The only restriction on an operation is that it takes as input a fixed number of elements of a set and returns an element of the same set. In this way, an operation can be thought of as a mapping on a set S , $m : S \times S \times \cdots \times S \rightarrow S$. The number of different elements taken as input, called operands, defines the "arity" of the operation. Most commonly studied are operations with arity 2, $m : S \times S \rightarrow S$, and these are called binary operations. An **identity** is an equation between two expressions, possibly involving variables, that is required to hold for every assignment of values to those variables in the underlying set (e.g. $a + 0 = a$ and $a + (-a) = 0$).

An **algebraic structure** consists of a nonempty set A , a collection of operations on A , and a collection of identities (known as axioms¹) that those operations satisfy. The following items pertain to algebraic structures:

Signature: a specification of the symbols used to describe a type of mathematical structure, together with the arity of each operation or relation. A signature may contain operation symbols, constant symbols (equivalently, 0-ary operation symbols), and relation symbols. A structure for a signature Σ consists of an underlying set together with an interpretation of each symbol in Σ . The signature specifies the *kind* of structure, but not the particular underlying set or the particular interpretations of its symbols. Loosely, **signature + interpretation + axioms = algebraic**

¹Specifically, non-logical axioms. Logical axioms are tautologies or logical truths. Non-logical axioms aim to capture what is special about a particular structure rather than what is forced by logic alone.

theory / structure: a structure tells you what operations exist, a particular structure / interpretation tells you what those operations actually are, and the axioms constrain those interpretations. For example, the signature of groups may be written

$$\Sigma_{\text{Grp}} = \left\{ \cdot^{(2)}, (\cdot)^{-1(1)}, e^{(0)} \right\},$$

corresponding to a binary operation, a unary operation, and a constant, respectively. A particular group G then interprets these operations in a specific way that aligns with the axioms (like, for $SO(3)$, the set is specifically the group comprises of the set of special, orthogonal matrices, the binary operation is matrix multiplication, the unary operation is the matrix inverse, and the 0-ary is the identity matrix).

- **Homomorphism:** a map between two structures of the same signature that preserves the interpretations of the signature's operations, constants, and relations (when present). In particular, if $f : A \rightarrow B$ is a map between two structures and g is an n -ary operation symbol in their common signature, then

$$f(g_A(x_1, \dots, x_n)) = g_B(f(x_1), \dots, f(x_n))$$

for all $x_1, \dots, x_n \in A$. Similarly, for a constant symbol c ,

$$f(c_A) = c_B.$$

One often says that f *preserves* the operations and constants.

- **Isomorphism:** a bijective homomorphism.
- **Endomorphism:** a homomorphism whose domain equals its codomain.
- **Automorphism:** an endomorphism that is also an isomorphism.

As an example, $\mathbb{R}^4$, the set of 4-dimensional tuples over $\mathbb{R}$ can be equipped with operations that can make it a vector space (vector addition and scalar multiplication in the normal way). $\mathcal{M}^{2 \times 2}$, the set of all 2×2 matrices over $\mathbb{R}$, can also be equipped with operations that make it a vector space (matrix addition and scalar multiplication in the normal way). They're both vector spaces, but they're different vector spaces. However, one can define an isomorphism between the two, showing that they share the same vector structure.

Chapter 2

Hierarchies of Mathematical Structures

Before we get into specific mathematical structures, it's helpful to note that many mathematical structures are more specific versions of other ones.

2.1 Algebraic

- **Magma:** A set with a binary operation (no requirement beyond closure).
- **Semigroup:** A magma but where the binary operation is associative.
- **Monoid:** A semigroup with an identity element (a "do nothing" operation).
- **Group:** A monoid with an additional inverse element for each element (for every element, there is an element that undoes that element under the action of the binary operation).
- **Abelian Group:** A group where the binary operation is also commutative.
- **Ring:** An abelian group with an additional binary operation (call it multiplication) that is associative and distributes over the abelian group's natural operation.
- **Commutative Ring:** A ring where the multiplicative operation is commutative.
- **Integral Domain:** A commutative ring with no zero divisors. I.e., if $ab = 0$, then $a = 0$ or $b = 0$ (this establishes that ring multiplication acts similarly to ordinary number multiplication, when it normally doesn't have to).
- **Field:** An integral domain with multiplicative inverses for all nonzero elements.

$$\text{Abelian Group} \subset \text{Group} \subset \text{Monoid} \subset \text{Semigroup} \subset \text{Magma}$$

$$\text{Field} \subset \text{Integral Domain} \subset \text{Ring} \subset \text{Abelian Group}.$$

Note that a ring is an example of an abelian group (in that it contains an abelian group within it), but it's better thought of as its own object: it's an abelian group with an entirely new operation equipped to it, and rings are usually thought of intuitively as a different object than a group.

- **Module:** an abelian group M and a ring R equipped with a map $*$: $R \times M \rightarrow M$ (often called scalar multiplication: the members of the group can be "multiplied" by the elements of R) subject to specific to compatibility conditions like

$$\begin{aligned} r(m_1 + m_2) &= rm_1 + rm_2, \\ (r_1 + r_2)m &= r_1m + r_2m, \\ (r_1r_2)m &= r_1(r_2m) \end{aligned}$$

- **Vector space:** a module where the ring is a field.
- **Algebra:** An algebra over a field F is a vector space over F with an additional operation $V \times V \rightarrow V$ (multiplication between vectors) such that

$$\begin{aligned} (a + b)c &= ac + bc, \\ a(b + c) &= ab + ac, \\ (\lambda a)b &= \lambda(ab) = a(\lambda b) \end{aligned}$$

Thus,

$$\text{Algebra} \subset \text{Vector Space} \subset \text{Module}.$$

2.2 Topological

- **Topological space:** a space equipped with enough structure to talk about which points are "near" each other in a qualitative sense, without specifying how near they are.
- **Metric space:** a space equipped with a notion of "distance" between each member of the set. The metric induces a natural topology (called the metric topology), so a metric space is a topological space.
- **Normed Vector space:** a vector space (see next chapter) equipped with a norm (a function that gives a notion of "length" for the vectors in the space). A norm induces a metric ($d(u, v) = ||u - v||$), so a normed vector space is a metric space is a topological space.
- **Inner Product space:** a vector space equipped with an inner product, which gives a notion of length and angles between vectors. An inner product induces a norm $||u|| = \sqrt{\langle u, u \rangle}$, so an inner product space is a normed vector space is a metric space is a topological space.
- **Banach space and Hilbert space:** a Banach space is a "complete" normed vector space and a Hilbert space is a "complete" inner product space. Completeness is a statement that the space contains all points that the space's own notion of convergence says should exist (more on this below).

So,

$$\begin{array}{ccccccc} \text{Inner Product space} & \subset & \text{Normed Vector Space} & \subset & \text{Metric Space} & \subset & \text{Topological Space} \\ & & \downarrow & & \downarrow & & \\ & & \text{Hilbert Space} & & \text{Banach Space} & & \end{array}$$

Lastly, we briefly cover manifolds (although these are just examples of topological spaces with extra structures, they're important).

- **Topological Manifold:** A topological space in which every member of the set locally resembles $\mathbb{R}^n$.
- **Smooth Manifold:** A manifold with a notion of differentiability. This allows us to define $T_p M$, the tangent space at p , which is a vector space.
- **Riemannian Manifold:** A smooth manifold such that every tangent space comes equipped with an inner product (the metric tensor). This allows us to define lengths and angles in a new, deformed space.

$$\text{Riemannian Manifold} \subset \text{Smooth Manifold} \subset \text{Topological Manifold}$$

Chapter 3

Topological Spaces

Definition 3.1 (Topological Space). *A topology on set X is defined as a collection τ of subsets of X (called open sets) satisfying:*

1. $\{0\}, X \in \tau$
2. Arbitrary unions of members of τ belongs to τ
3. Finite intersections of members of τ belongs to τ

A **neighborhood** of a point $x \in X$ is any open set containing x (some define a neighborhood of x as any set that contains an open set that contains x , but this feels unintuitive to me).

Definition 3.2 (Continuous function). *A function $f : X \rightarrow Y$ between topological spaces $(X, \tau_X), (Y, \tau_Y)$ is called continuous at $x \in X$ if for every neighborhood $N \in \tau_Y$ of $f(x)$, there is a neighborhood $M \in \tau_X$ of x such that $f(M) \subseteq N$. If f is continuous $\forall x \in X$, then f is simply called continuous.*

The idea: similarly to how continuity is defined over the real line, no matter how small a neighborhood W around $f(x)$, we choose, we can always find a neighborhood around x , U , such that $f(U) \subset W$. In other words, we can always "get closer" to the point $f(x)$ by "getting closer" to point x .

Definition 3.3 (Homeomorphism). *A function $f : X \rightarrow Y$ between two topological spaces is called a homeomorphism if:*

1. f is a bijection
2. f is continuous
3. f^{-1} is continuous

A subset $A \subseteq X$ is closed in X if it's complement is open. Equally, this means that A is equal to its closure (the closure being the set itself plus all of it's boundary points). A subset $A \subseteq X$ is said to be a dense subset of X if the closure of A is equal to X . This means that any point in X can be arbitrarily-well approximated by points in A .

3.1 Metric Topology

A metric space is a set M with a binary operation $d : M \times M \rightarrow \mathbb{F}$ defined such that

1. $d(x, x) = 0 \forall x \in M$.

- 2. Positivity: If $x \neq y$, then $d(x, y) > 0$.
- 3. Symmetry: $d(x, y) = d(y, x)$.
- 4. Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Definition 3.4 (Ball). *A metric ball of radius $r > 0$ centered at point $p \in M$ is defined as $B_r(p) = \{x \in M \mid d(x, p) < r\}$.*

This induces a topology on the metric space (M, d) , τ_d , defined such that $U \in \tau_d$ if $\forall u \in U, \exists B_r(u) \subseteq U$ for some $r > 0$. This is called the metric topology.

3.2 Manifolds

Chapter 4

Vector Spaces

4.1 Fields

Field: A non-empty set F equipped with two binary operations, addition (+) and multiplication ($\cdot$) satisfying the following axioms:

- Associativity of addition and multiplication: $a + (b + c) = (a + b) + c$, and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- Commutativity of addition and multiplication: $a + b = b + a$, and $a \cdot b = b \cdot a$.
- Additive and multiplicative identity: $\exists$ elements $0, 1 \in F$ s.t. $a + 0 = a$ and $a \cdot 1 = a \ \forall a \in F$.
- Additive inverses: $\forall a \in F, \exists -a \in F$ such that $a + (-a) = 0$.
- Multiplicative inverses: $\forall a \in F \setminus \{0\}, \exists a^{-1} \in F$ (also labeled $1/a \in F$) s.t. $a \cdot a^{-1} = 1$.
- Distributivity of multiplication over addition: $a \cdot (b + c) = a \cdot b + a \cdot c$.

The signature of a field is given by

$$\Sigma_{\text{field}} = \{.,^{(2)}, +^{(2)}, 0^{(0)}, 1^{(0)}\}.$$

This signature only includes primitive operations. Another commonly used definition of the signature could include the induced operations (the inverses as unary operations), which would look like

$$\Sigma_{\text{field}} = \{.,^{(2)}, +^{(2)}, 0^{(0)}, 1^{(0)}, -^{(1)}, {}^{-1}{}^{(1)}\},$$

where it's important to note that ${}^{-1} : F \setminus \{0\} \rightarrow F \setminus \{0\}$. We avoid using definition as, to us, the signature is there to specify the primitive operations of the language. Operations that are uniquely determined by these operations and the axioms need not be included as primitive operations, and including them can often complicate things. For example, there's the added issue that the multiplicative operation ${}^{-1}$ is *technically* not a unary operation on F as it's domain and image exclude the additive identity element $\{0\} \in F$.

However, the motivation for the richer signature definition is as follows: the axioms are part of the structure, so operations implied by the axioms characterize the structure just as much as primitive operations do; as such, the signature should maybe include these operations.

Vector space: A vector space over a field F is a non-empty set V equipped with a binary operation $+: V \times V \rightarrow V$ (vector addition) and a binary function $\cdot : F \times V \rightarrow V$ (scalar multiplication, also

often written as $\cdot(a, v) = a \cdot v = av$ for $a \in F$ and $v \in V$) that satisfy the following eight axioms (in the following, $a, b \in F$, and $u, v, w \in V$, and, unless otherwise stated, the following apply to all such elements in the sets):

- Associativity of vector addition: $u + (v + w) = (u + v) + w$.
- Commutativity of vector addition: $u + v = v + u$.
- Identity element of vector addition: $\exists 0 \in V$, called the zero vector, s.t. $v + 0 = v \ \forall v \in V$.
- Inverse elements of vector addition: $\forall v \in V, \exists -v \in V$ s.t. $v + (-v) = 0$.
- Compatibility of scalar multiplication with field multiplication: $a \cdot (bv) = (a \cdot_F b)v$, where $\cdot_F$ is the multiplication operation in the field F .
- Identity element of scalar multiplication: $1_F v = v$, where 1_F is the multiplicative identity in the field F .
- Distributivity of scalar multiplication with respect to vector addition: $a(u + v) = au + av$
- Distributivity of scalar multiplication with respect to field addition: $(a +_F b)v = av + bv$, where $+_F$ is the addition operation in the field F .

A vector space over a field can be written as $(V, F, +_V, \cdot_V)$ where V is the set of vectors, F is the field, $+$ is the vector addition operation, and $\cdot$ is the scalar multiplication binary function. However, unless being explicit is necessary, the vector space is often just written as V with the field and operations implied. Vector spaces are, in some sense, complicated structures as they involve the coupling of two algebraic structures: the set of vectors with the vector addition operation forms an abelian group under vector addition (more on this below), while the scalars form a field F under their own addition and multiplication. Scalar multiplication $F \times V \rightarrow V$ couples these structures through the above axioms. For a fixed field F , a vector space can nevertheless be described by a signature in which scalar multiplication is represented by a family of unary operations $v \mapsto av$, one for each $a \in F$. In this way, the signature for a vector space over a fixed field F could be written as

$$\Sigma_{(V,F)} = \{+_V^{(2)}, 0_V^{(0)}, \{\lambda^{(1)} : \lambda \in F\}\}.$$

Here, we've neglected the implied unary operations and only included primitive operations. The $\{\lambda^{(1)} : \lambda \in F\}$ is saying that for ever fixed scalar $\lambda \in F$, there is a unary operation $\lambda_V : V \rightarrow V$, $v \mapsto \lambda v$. Further, we treat the fixed field F as external, which is why this signature neglects the operations defining the field F itself. If one wanted the signature to instead represent the entire coupled structure, rather than treating F as fixed external data, one could instead use a two-sorted signature containing the field operations on F , the vector space operations on V , and the scalar multiplication $F \times V \rightarrow V$. We neglect to do this as, for our purposes, this level of detail is pedantic and has diminishing returns.

4.2 General Properties of Vector Spaces

Let V be a vector space over field F .

- Linear combination: a linear combination of a set of vectors $\{v_1, v_2, \dots, v_n\}$ in V is a vector of the form $a_1 v_1 + a_2 v_2 + \dots + a_n v_n$ for some $a_1, a_2, \dots, a_n \in F$.

- Subspace of V : if V is a vector space over field F , a subset $U \subset V$ is a subspace of V if U is also a vector space over F for the operations of V .
- Span of subset $U \subset V$: the intersection W of all subspaces of V that contain U : the smallest subset of V containing U . Identically, the span is the set of all possible linear combinations of the vectors in U : when $U = \{v_1, \dots, v_n\}$

$$\text{span}(U) = \{a_1v_1 + a_2v_2 + \dots + a_nv_n \mid a_1, \dots, a_n \in F\}.$$

- A subset of vectors $\{v_1, v_2, \dots, v_n\}$ is called linearly independent if a linear combination of all vectors in the subset is only zero iff the coefficients in the linear combination are all identically zero:

$$\sum_{i=1}^n a_i v_i = 0$$

implies all $a_i = 0$. Otherwise, the vectors are called linearly dependent. A subset $U \subset V$ is called linearly independent if all vectors in the subset are linearly independent with respect to each other.

- Finite-dimensional vector space: a vector space V such that $\exists$ a finite subset $U \subset V$ for which $\text{span}(U) = V$. This implies that, $\forall v \in V$, $\exists a_1, a_2, \dots, a_n \in F$ and $v_1, v_2, \dots, v_n \in V$ s.t.

$$v = \sum_{i=1}^n a_i v_i$$

for n finite.

- Basis: a linearly independent subset of vectors that spans the space. A basis is guaranteed to exist for a finite-dimensional vector space, and every such basis has the same number of elements. An ordered basis is a basis $\{e_1, e_2, \dots, e_n\}$ of V such that the order of those elements is maintained (this is only important for consistency when discussing certain linear algebra topics like basis transformations and not something that changes the math).
- Dimension: the number of vectors in any basis of a finite-dimensional vector space.
- For a finite-dimensional vector space of dimension n , any subset with greater than n elements must be linearly dependent (but *can* span the space), and any subset with less than n elements must not span the space (but *can* be linearly independent).
- If two spaces have the same dimension, then there is necessarily an isomorphism uniting them, and they're said to be isomorphic.
- There can exist a notion of a basis in infinite-dimensional vector spaces, but many pleasant theorems assume we're working with a finite-dimensional space.
- Direct sum: if $W, U \subseteq V$ are subspaces of vector space V , their sum is the subspace $W + U = \{w + u \mid w \in W \text{ and } u \in U\}$. If $W \cap U = \{0\}$, then the sum is called a direct sum and is denoted $W \oplus U$. In that case, every vector in $W \oplus U$ can be uniquely written as a sum $w + u$ with $w \in W$, $u \in U$ unique. If $V = W \oplus U$, then $W \oplus U$ is a direct sum decomposition of V .

4.3 Linear Maps

Let V, W, X be vector spaces over a field F .

- **Linear map:** A map $T : V \rightarrow W$ such that

$$T(av + bu) = aT(v) + bT(u)$$

$\forall a, b \in F, v, u \in V$. Any such map can be represented as a matrix.

- **Bilinear map:** A map $T : V \times W \rightarrow X$ that is linear in each argument separately:

$$T(av_1 + bv_2, u) = aT(v_1, u) + bT(v_2, u) \text{ and}$$

$$T(v, cu_1 + du_2) = cT(v, u_1) + dT(v, u_2)$$

$\forall a, b, c, d \in F, u_1, u_2 \in W, v_1, v_2 \in V$.

- **Multilinear map:** A map $T : V_1 \times V_2 \times \cdots \times V_n \rightarrow W$ that is linear in each argument separately.

- **Sesquilinear:** A map $T : V \times W \rightarrow F$ that is linear in one argument and conjugate-linear in the other. For complex vector spaces (say, over the field $\mathbb{C}$):

$$T(av_1 + bv_2, u) = aT(v_1, u) + bT(v_2, u) \text{ and}$$

$$T(v, cu_1 + cu_2) = c^*T(v, u_1) + d^*T(v, u_2)$$

$\forall a, b, c, d \in F, v_1, v_2 \in V, u_1, u_2 \in W$.

Like mentioned above, any linear map can be represented as a matrix and bilinear maps can also be represented as a matrix. Let V be a finite-dimensional vector space of dimension n with a basis $E = \{e_1, e_2, \dots, e_n\}$. Then, any vector $v \in V$ can be written as $v = \sum_{i=1}^n a_i e_i$ for some $a_i \in F$. The $(n, 1)$ matrix

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

is called a column vector or the coordinate vector of v in the basis E . Sometimes, this is written as $[v]_E$, though sometimes it's understood just by context when writing v . What this is really hiding is an isomorphism $\phi_E : F^n \rightarrow V$ (an n -dimensional vector space is necessarily isomorphic to $F^n = F \times F \times \cdots \times F$ with the standard definitions of vector addition and scalar multiplication implied by the Cartesian product of the field):

$$\phi_E(a_1, \dots, a_n) = \sum_i a_i e_i.$$

Let E be a basis for V and let E' be a basis for another vector space W of dimension m . Then, the action of the linear map $T : V \rightarrow W$ can be expressed as

$$T(v) = w \rightarrow T\left(\sum_{i=1}^n a_i e_i\right) = w \tag{4.1}$$

$$\sum_{i=1}^n a_i T(e_i) = w. \tag{4.2}$$

So, the action of the transformation is completely determined by how it maps the basis elements of one space to the other space. Thus, expanding $T(e_i) \in W$ in the basis of E' , we find

$$T(e_i) = \sum_{j=1}^m T_{ji} e'_j \quad (4.3)$$

for some $T_{ji} \in F$. Thus,

$$T(v) = \sum_{i=1}^n a_i \sum_{j=1}^m T_{ji} e'_j = \sum_{j=1}^m e'_j \left(\sum_{i=1}^n T_{ji} a_i \right). \quad (4.4)$$

One can write this in matrix form as

$$[T(v)]_{E'} = \begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22} & \cdots & T_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ T_{m1} & T_{m2} & \cdots & T_{mn} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}. \quad (4.5)$$

From this, it's clear that the matrix

$$\begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22} & \cdots & T_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ T_{m1} & T_{m2} & \cdots & T_{mn} \end{pmatrix}$$

is the matrix form of the transformation in the basis E and E' $[T]_{E \rightarrow E'}$, and the columns of this matrix $[T(e_i)]_{E'}$.

One can see this more formally by treating E and E' as tuples: $(E) = (e_1, \dots, e_n)$ and $(E') = (e'_1, \dots, e'_m)$ and extending T component-wise across the tuple: $TE := (T(e_1), \dots, T(e_n))$. Using the coordinate isomorphism associated with E' : $\phi_{E'} : F^m \rightarrow W$, we can extend the inverse across the tuple TE :

$$\phi_{E'}^{-1}(TE) = ([T(e_1)]_{E'}, \dots, [T(e_n)]_{E'}).$$

Technically, this is an n -tuple of m -tuples: $(F^m)^n$. To be maximally formal, we can introduce the canonical isomorphism $\sigma : (F^m)^n \rightarrow F^{m \times n}$ s.t.

$$(v_1, \dots, v_n) \mapsto \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix}.$$

This immediately implies the matrix we established $[T]_{E \rightarrow E'} = (\sigma \circ \phi_{E'}^{-1})(TE)$.

Here are some general properties of linear maps:

- Kernel: $\ker T = \{v \in V : T(v) = 0\}$.
- Image: $\operatorname{im} T = \{T(v) : v \in V\} \subseteq W$
 - $\ker T \subseteq V$ is a subspace of V and $\operatorname{im} T \subseteq W$ is a subspace of W .
 - $\operatorname{nullity}(T) = \dim \ker T$ and $\operatorname{rank}(T) = \dim \operatorname{im} T$

- $\dim V = \text{nullity}(T) + \text{rank}(T)$.
- T injective if $T(v_1) = T(v_2)$ implies $v_1 = v_2$, which means $\ker T = \{0\}$
- T is surjective if $\text{im } T = W$
- If T is surjective, $\text{im } T = W$, so $\dim V = \text{nullity } T + \dim W$. If T is injective, then $\text{nullity } T = 0$. So, if T is bijective, $\dim V = \dim W = \text{rank}(T)$, and T is called full-rank. This establishes an isomorphism between V and W . Like any bijection, T must then be invertible.
- If T is not a bijection, then T "kills" off dimensions. For example, T might take a vector in $\mathbb{R}^3$ and project it to a specific 2D plane embedded in $\mathbb{R}^3$.
- If T is an isomorphic endomorphism, $T : V \rightarrow V$, then T can be thought of as a change-of-basis transformation: a transformation that leaves the fundamental vector unchanged and just changes the basis it's represented in. This is just an interpretation. T can be thought of as active: a transformation actively changing the input vector, or a passive transformation: a transformation that leaves the vector unchanged but transforms the basis it's represented in. If the transformation isn't full-rank, this
 - Eigenspace: The λ -eigenspace of T is the set $E_\lambda = \{v \in V \mid T(v) = \lambda v\}$ or, identically, $E_\lambda = \ker(T - \lambda I)$.
 - Determinant: T is invertible iff $\det T \neq 0$ (which implies $\text{rank } T = n$, so the determinant detects whether an endomorphism loses a dimension).

4.4 Eigenvectors

4.5 Determinant

Chapter 5

Dual Spaces

Let V be a vector space over field $\mathbb{F}$. The dual space of V , V^* , is the space of all linear functionals $f : V \rightarrow \mathbb{F}$.

Chapter 6

Inner Product Spaces

An inner product space is a vector space V over a field $\mathbb{F} = \mathbb{R}, \mathbb{C}$ equipped with a binary operation $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$, called the inner product, satisfying the following properties:

1. Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
2. Linearity in the first argument: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.
3. Positive-definiteness: if $x \neq 0$, then $\langle x, x \rangle > 0$.

A few basic properties of the inner product:

- $\langle 0, x \rangle = \langle x, 0 \rangle = 0$.
- $\langle x, x \rangle$ is real and non-negative.
- $\langle x, x \rangle = 0$ iff $x = 0$.
- $\langle x, ay + bz \rangle = \bar{a}\langle x, y \rangle + \bar{b}\langle x, z \rangle$ (conjugate-linearity in the second argument).

Since $\langle \cdot, \cdot \rangle$ is linear in the first argument and conjugate-linear in the second argument (as the last property shows), the inner product is a sesquilinear form. If $\mathbb{F} = \mathbb{R}$, conjugate symmetry reduces to symmetry, and sesquilinear reduces to bilinear. Occasionally, it's useful to relax the positive-definite criterion to positive-semidefinite: $\langle x, x \rangle \geq 0$. This defines a positive, semi-definite Hermitian form. The inner product induces a norm $\| \cdot \|_V : V \rightarrow \mathbb{R}$ via $\|v\|_V = \sqrt{\langle v, v \rangle}$. This norm induces a metric $d_V : V \times V \rightarrow \mathbb{R}$ via $d_V(v, u) = \|v - u\|_V$. This metric induces a topology, called the metric topology.

6.0.1 Projectors

$P : V \rightarrow V$ such that $P^2 = P$ (idempotent). Then, $V = \text{Im}(P) \oplus \text{Ker}(P)$, $P|_{\text{Im}(P)} = I_{\text{Im}(P)}$. Orthogonal projector: a projector such that $\text{Im}(P)$ and $\text{Ker}(P)$ are orthogonal subspaces.

6.0.2 Isometries

Map that preserves inner products (more generally, maps that preserve the metric). For an inner product space, T is an isometry if $\langle Tx, Tx \rangle = \langle x, x \rangle \forall x \in V$. This forces $\langle Tx, Ty \rangle = \langle x, y \rangle \forall x, y \in V$. This forces $T^\dagger T = I$. Notably, unless $\text{Im}(T) = V$, $TT^\dagger \neq I$ (the isometry need be

injective, but it need not be surjective). It can be shown that $TT^\dagger$ is an orthogonal projector onto $\text{Im}(T)$.

6.1 Hilbert and Banach spaces

6.1.1 Finite Dimensions

First, we discuss finite dimensional Hilbert spaces and the theorems that follow.

Theorem 6.1.1 (Finite dimensional inner product spaces are Hilbert spaces). *Every finite dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$ is a Hilbert space (under the standard induced topology).*

Theorem 6.1.2 (Adjoint Exists and is Unique). *For any operator A on a finite dimensional Hilbert space $\mathcal{H}$, there exists a unique operator $A^\dagger$, called the adjoint, such that $\langle y, Ax \rangle = \langle A^\dagger y, x \rangle \forall x, y \in \mathcal{H}$ as long as the inner product is non-degenerate.*

Theorem 6.1.3 (Spectral Theorem). *Let $\mathcal{H}$ be a finite-dimensional Hilbert space over $\mathbb{C}$ with dimension n . Every Hermitian A can be written as $A = UDU^\dagger$ with U unitary and D real diagonal. Equivalently, $A = \sum_i \lambda_i P_i$ with real λ_i and mutually orthogonal projections P_i that sum to $\mathbb{I}$.*

Lemma 6.1.3.1. *An eigenvalue for A exists.*

Proof. The characteristic polynomial $\det(A - \lambda \mathbb{I})$ has degree $m \geq 1$. By the fundamental theorem of algebra, it must have at least one root $\lambda \in \mathbb{C}$. At that root, $A - \lambda \mathbb{I}$ is singular, so $\ker B \neq \{0\}$. I.e., some $v \neq 0$ satisfies $Av = \lambda v$. $\square$

Lemma 6.1.3.2. *Eigenvalues of A are real.*

Proof. Let v be an eigenvector of A with corresponding eigenvalue λ . Then, $\langle v, Av \rangle = \lambda \langle v, v \rangle$. Since A is Hermitian, $\langle v, Av \rangle = \langle A^\dagger v, v \rangle = \langle Av, v \rangle = \lambda^* \langle v, v \rangle$. So, $\lambda \langle v, v \rangle = \lambda^* \langle v, v \rangle$. $\langle v, v \rangle$ is only equal to zero if $v = 0$ and $v \neq 0$ by definition of an eigenvector, so $\langle v, v \rangle \neq 0$ and thus $\lambda^* = \lambda$. Thus, $\lambda \in \mathbb{R}$. $\square$

Lemma 6.1.3.3. *The orthogonal complement of a subspace W is the set of all vectors that are orthogonal to every single vector in W . It's often denoted $W^\perp$. The orthogonal complement of an eigenspace is invariant under A . Let w be a single eigenvector. Then, A restricted to $w^\perp$ is again Hermitian, now on a space of dimension $n - 1$.*

Proof. Let V the eigenspace of A associated with λ . Then, if $w \in V^\perp$ and $v \in V$, $\langle v, w \rangle = 0$. A is Hermitian, so $\langle v, Aw \rangle = \langle Av, w \rangle = \lambda \langle v, w \rangle = 0$. So, if $w \in V^\perp$, then $Aw \in V^\perp$. In other words, the orthogonal complement of an eigenspace is invariant under the action of A .

Let $v \neq 0$ and consider the linear functional defined by $\phi(x) := \langle v, x \rangle \forall x \in \mathcal{H}$. Then, $v^\perp = \ker \phi$, by definition. ϕ is not the zero functional because $\phi(v) = \|v\|^2 \neq 0$. Thus, the range of ϕ is a nonzero subspace of $\mathbb{C}$ (which must of course be all of $\mathbb{C}$) and $\dim \text{Ran } \phi = 1$. By rank-nullity theorem, $\text{rank } \phi + \text{nullity } \phi = n$, so $\dim \ker \phi = \dim v^\perp = n - 1$ (any non-zero linear functional has a rank = 1 and $\dim \ker \phi = n - 1$). Basically, this statement is just saying that a linear functional collapses a vector space with dimension n to one with dimension 1, so, as long as it's nonzero, it must kill off $n - 1$ dimensions). Thus, $v^\perp$ is an $n - 1$ -dimensional subspace of $\mathcal{H}$.

$A|_{v^\perp}$, well defined since A maps $v^\perp$ to $v^\perp$, is of course still Hermitian: $\langle y, Ax \rangle = \langle Ay, x \rangle \forall x, y \in \mathcal{H}$, so $\langle u, A|_{v^\perp} w \rangle = \langle A|_{v^\perp} u, w \rangle \forall u, w \in v^\perp \subset \mathcal{H}$. $\square$

Now, we induct.

Statement $P(n)$: for every complex inner product space V of dimension n , and every Hermitian $A : V \rightarrow V$, there is an orthonormal basis of V consisting of eigenvectors of A , and all of the eigenvalues are real.

Base case ($n = 1$): Pick any unit vector e . Since $V = \text{span } e$, $Ae = ce$ for some $c \in \mathbb{C}$. So, e is an eigenvector and $\{e\}$ is an orthonormal basis. Hermiticity guarantees c real.

Assume $P(n-1)$, prove $P(n)$: Let $\dim V = n$ and let A be Hermitian on V . By the first lemma, A has at least one eigenpair, (v, λ) . If it has multiple, choose one. Then normalize $e_1 := v/\|v\|$. By the second lemma, $\lambda \in \mathbb{R}$. Now, split the space: let $W := e_1^\perp$. Then, $V = \text{span}(e_1) \oplus W$, and $\dim W = n-1$. Equip W with the inner product inherited from V . Then, W is an $(n-1)$ -dimensional inner product space. By the third lemma, $A_W := A|_W : W \rightarrow W$, the restriction of A onto W , is a well-defined operator (it maps W to W), and it's Hermitian on W . Since $P(n-1)$, W has an orthonormal basis $\{e_2, \dots, e_n\}$ with $A_W e_j = \lambda_j e_j$ and $\lambda_j \in \mathbb{R}$. Since $A_W e_j = A e_j$, these are eigenvectors of A itself. $e_2, \dots, e_n$ are orthonormal and each lie in $W = e_1^\perp$, so each is orthogonal to e_1 . Thus, $e_1, e_2, \dots, e_n$ are orthonormal, meaning they're linearly independent. $\dim V = n$, and $e_1, e_2, \dots, e_n$ form n , linearly-independent vectors in V . Thus, $\{e_1, e_2, \dots, e_n\}$ form an orthonormal basis in V and $A e_i = \lambda e_i$ with $\lambda \in \mathbb{R}$. This proves $P(n)$.

$V = \bigoplus_{i=1}^n \text{span}(e_i)$. Place e_i as the columns of U . Then, U is unitary, and $A = U D U^\dagger$ with $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ (note that nothing is currently forcing the eigenvalues to be distinct). Define the orthogonal projector $P_k := \sum_{i:\lambda_i=\mu_k} |e_i\rangle\langle e_i|$. Then, $A = \sum_k \mu_k P_k$. Lastly, let $E_k := \ker(A - I\mu_k) = \text{Ran } P_k$. Then, $V = \bigoplus_{i=1} E_i$. This can be seen by noting that $E_k = \bigoplus_{i:\lambda_i=\mu_k} \text{span}(e_i)$. And, since distinct eigenvalues result in orthogonal eigenvectors, $E_k \cap E_{k'} = \{0\}$ for $k \neq k'$.

We can write this in measure notation using the following. Take M to be the Borel sets of $\mathbb{R}$. Then, define $E := E(\Delta) = \sum_{i:\lambda_i \in \Delta} P_i \ \forall \ \Delta \in M$. V is finite-dimensional, so it's necessarily separable. All finite-dimensional operators are bounded, so $E(\Delta)$ is bounded. $E(\Delta)$ is a sum of mutually orthogonal projections, so $E(\Delta)$ is an orthogonal projection. $E(\Delta)$ is a sum of self-adjoint operators, so $E(\Delta)$ is a self-adjoint operator. $E(0) = 0$ since no eigenvalue is zero. $E(\mathbb{R}) = \sum_i \lambda_i P_i = I$ since V is a direct sum of the eigenspaces. If $E_1, E_2, \dots$ in M are disjoint, then $\forall v \in V$,

$$E\left(\tilde{E} = \bigcup_{j=1}^{\infty} E_j\right) v = \left(\sum_{j:\lambda_j \in \tilde{E}} P_j\right) v = \sum_{j=1}^{\infty} \sum_{k:\lambda_k \in E_j} P_k v = \sum_{j=1}^{\infty} E(E_j) v.$$

Thus, E is a projection-valued measure on V . Define $\lambda : \mathbb{R} \rightarrow \mathbb{R}$ such that $\lambda = \sum_i \lambda_i I_{\lambda_i}$ (that is, a sum of indicator functions). λ is a measurable function, and

$$A = \int_{\mathbb{R}} \lambda dE = \sum_i \lambda_i P_i.$$

6.1.2 Infinite Dimensions

6.1.3 Spectral Theorem

If A is a densely-defined, bounded, self-adjoint operator on a separable Hilbert space $\mathcal{H}$, then $\exists$ a unique projection-valued measure E on the Borel σ -algebra of $\mathbb{R}$ such that

$$A = \int_{\mathbb{R}} \lambda dE(\lambda).$$

Chapter 7

Tensor Spaces

Cartesian product: the Cartesian product of sets A and B is the set denoted $A \times B$ including all ordered pairs (a, b) where $a \in A$ and $b \in B$: $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$. **Tensor product:** the tensor product of vector spaces V and W (both over the field $\mathbb{F}$) is the vector space, denoted $V \otimes W$, wherein the set of elements is the Cartesian product $V \times W$, and vector addition are defined according to V and W , applied component-wise. In other words, there is a bilinear map $V \times W \rightarrow V \otimes W$ that maps (v, w) ($v \in V, w \in W$) to an element $v \otimes w \in V \otimes W$. This is guaranteed to be a vector space, by construction. We often look at tensors that are combinations of a vector space and its dual, like

$$\mathcal{T} = V^* \otimes \cdots \otimes V^* \otimes V \otimes \cdots \otimes V^*$$

If there are p copies of V^* and q copies of V , then $T \in \mathcal{T}$ is called a type (p, q) tensor. There is a one-to-one correspondence between tensors thought of in this way and tensors thought of as linear maps:

$$T = V^* \times \cdots \times V^* \times V \times \cdots \times V \rightarrow \mathbb{F}.$$

7.1 Transformation Laws

7.2 Examples of Tensors

Linear transformations are type $(1, 1)$ tensors. Bilinear forms are type $(0, 2)$ tensors.

7.3 Adding a Dual Correspondence

7.3.1 Metric Tensor

Chapter 8

Groups

Definition 8.1 (Group). *A group is a set G with a binary operation on G , $\cdot : G \times G \rightarrow G$ such that*

1. *Associativity: $\forall a, b, c \in G, (a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
2. *Identity element: $\exists e \in G$ s.t. $\forall g \in G, e \cdot g = g$ and $g \cdot e = g$. This element is unique.*
3. *Inverse element: $\forall g \in G, \exists g' \in G$ s.t. $g \cdot g' = e$ and $g' \cdot g = e$ where e is the identity element. For each $g \in G$, the element g' is unique. It's called the inverse of g and is often denoted g^{-1} .*

Definition 8.2 (Group Homomorphism). *Given two groups $(G, *)$ and $(H, \cdot)$, a group homomorphism from $(G, *)$ to $(H, \cdot)$ is a function $h : G \rightarrow H$ s.t.*

$$h(u * v) = h(u) \cdot h(v) \quad \forall u, v \in G.$$

From this, it can be deduced that $h(e_G) = e_H$ and $h(u^{-1}) = h(u)^{-1}$, so h is "compatible with the group structure", just as every homomorphism should be compatible with the underlying structure.

8.1 Representations

Definition 8.3 (Representation). *A representation of a group G on a vector space V over field $\mathbb{F}$ is a group homomorphism $\rho : G \rightarrow GL(V)$ ($GL(V)$ is the group of all automorphisms on V , with functional composition as the group operation). I.e.,*

$$\rho(g_1 g_2) = \rho(g_1) \rho(g_2) \quad \forall g_1, g_2 \in G.$$

For a representation ρ over vector space V , a subspace $W \subset V$ is called G -invariant if $\rho(g)W \subseteq W$ $\forall g \in G$.

Definition 8.4 (Irreducible Representation). *A representation ρ over vector space V is called irreducible if the only invariant subspaces are $W = \{0\}$ and $W = V$.*

8.2 Lie Groups

Chapter 9

Measures

Measures and measure spaces are designed to generalize the notion of "size" or "volume" for sets. A σ -algebra of a set X is a subset of $P(X)$ that is suitably measurable.

Definition 9.1 (σ -algebra). *Let X be a set and let $P(X)$ be its power set (the set of all subsets of X). Then, a subset $\Sigma \subseteq P(X)$ is called a σ -algebra if*

1. $X \in \Sigma$
2. Σ is closed under complementation.
3. Σ is closed under countable unions.

The idea being: if we can measure the "size" of $W \subset X$, we should immediately know the "size" of $X \setminus W$. Further, if we know the "size" of a number of subsets, we should be able to tell the "size" of them added together. Notably, we'd like to be able to find the "size" of countably infinite measurable sets added together. These axioms then imply $\emptyset \in \Sigma$ and closure under countable intersections. Note: this definition is very similar to that of a topology, except topologies don't care about complements at all. A notion of "closeness" doesn't really restrict how the complement behaves (in a direct way), but a notion of a "measurable set" certainly does.

Definition 9.2 (Borel σ -algebra). *The Borel σ -algebra over a topological space X is the smallest σ -algebra one can construct that contains all the open sets defined by the topology.*

Now, we want a way to "measure the size" of sets. We do this with a function called the measure.

Definition 9.3 (Measure). *Let X be a set and Σ a σ -algebra over X . A function $\mu : \Sigma \rightarrow [0, +\infty]$ ($[0, +\infty]$ is the set of non-negative real numbers together with an element $+\infty$ that is used to denote elements greater than all other (so-called finite) elements) is called a measure if*

1. $\mu(\emptyset) = 0$
2. Countable additivity: $\forall$ countable collections $E_{k=1}^{\infty}$ of pairwise disjoint sets in Σ ,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

The pair (X, Σ) is called a measurable space, and the triple (X, Σ, μ) is called a measure space.

9.1 Lebesgue Integration

With this new notion of a "measure", we can discuss integration of a function $f : X \rightarrow Y$ over a set X :

$$\int_X f d\mu_X.$$

9.1.1 Indicator Functions

9.1.2 Non-negative Simple Functions

9.1.3 General Non-negative Functions

9.1.4 Projection Valued Measure

A topological space X is separable if it contains a countable dense subset. Take U to be this countable dense subset of X . Then, any $x \in X$ can be arbitrarily well approximated by a countable series in U . Let $\mathcal{H}$ be a separable complex Hilbert space (separability is vital for things like Fourier expansions). Let (X, M) be a measurable space consisting of a set X and a Borel σ -algebra M on X . A projection valued measure π is a map from M to the set of bounded (bounded: $\exists C > 0$ such that $\forall v \in V, \|Av\| \leq C\|v\|$, but it implies continuity for a separable Hilbert space) self-adjoint operators on $\mathcal{H}$ such that

1. $\pi(E)$ is an orthogonal projection $\forall E \in M$.
2. $\pi(\emptyset) = 0$ and $\pi(X) = I$.
3. If $E_1, E_2, \dots$ in M are disjoint, then $\forall v \in \mathcal{H}$,

$$\pi\left(\bigcup_{j=1}^{\infty} E_j\right)v = \sum_{j=1}^{\infty} \pi(E_j)v$$

Part II

Physics Foundations

Chapter 10

Quantum Mechanics

10.1 Deriving the TISE from the TDSE

The time evolution operator obeys the time-dependent Schrödinger (TDSE):

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H} \hat{U}(t, t_0). \quad (10.1)$$

In the cases where the Hamiltonian is time-independent, as is our case,

$$\hat{U}(t, t_0) = \exp \left(\frac{-i\hat{H}(t - t_0)}{\hbar} \right). \quad (10.2)$$

Acting on an energy eigenket $|E\rangle$ such that $\hat{H}|E\rangle = E|E\rangle$,

$$|E\rangle_t = \hat{U}(t, t_0) |E\rangle = \exp \left(-\frac{i}{\hbar} E(t - t_0) \right) |E\rangle_{t_0}.$$

Such a state only varies in phase as time progresses. It remains an eigenket of the Hamiltonian with energy E for all time. Using completeness of the energy eigenkets, any ket can be written as $|\Psi\rangle = \sum_E |E\rangle \langle E|\Psi\rangle$, and so the time dependence of any arbitrary state is:

$$|\Psi\rangle_t = \sum_E \exp \left(-\frac{i}{\hbar} E(t - t_0) \right) \langle E|\Psi\rangle_{t_0} |E\rangle.$$

By applying a ket at t_0 on the right of both sides of eq. (10.1), we can derive the more conventional form of the Time-dependent Schrödinger equation (TDSE):

$$i\hbar \partial_t |\Psi\rangle_t = \hat{H} |\Psi\rangle_t. \quad (10.3)$$

Now, we restrict eq. (10.3) to a single energy eigenspace. In that eigenspace, the eigenket evolves as $|E\rangle_t = \phi(t) |E\rangle_{t_0}$, where $\phi(t) = \exp(-iE(t - t_0)/\hbar)$ as discussed earlier. Thus,

$$\begin{aligned} i\hbar \partial_t (\phi(t) |E\rangle_{t_0}) &= \hat{H} (\phi(t) |E\rangle_{t_0}) \\ i\hbar (-iE/\hbar) \phi(t) |E\rangle_{t_0} &= \phi(t) \hat{H} |E\rangle_{t_0} \\ E \phi(t) |E\rangle_{t_0} &= \phi(t) \hat{H} |E\rangle_{t_0} \\ E |E\rangle_{t_0} &= \hat{H} |E\rangle_{t_0}. \end{aligned}$$

This approach derives the time-independent Schrödinger equation (TISE) from abstract operator algebra, which is more general than the standard approach: starting with the TDSE and using the position basis, the assumption that H doesn't depend on t , and separation of variables to arrive at the same equation.

10.2 Separating out the center of mass

Consider a two-particle system where the particles have masses m_1 and m_2 and the particles' positions are $\mathbf{r}_1$ and $\mathbf{r}_2$. We can define the center of mass and relative coordinate as

$$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2} \quad \mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2,$$

respectively. Then, $\frac{\partial}{\partial r_{ij}} = \frac{\partial r_k}{\partial r_{ij}} \frac{\partial}{\partial r_k} + \frac{\partial R_k}{\partial r_{ij}} \frac{\partial}{\partial R_k}$ (where r_{ij} is the j -th coordinate of the i -th particle, and Einstein summation convention has been adopted). Thus,

$$\begin{aligned} \frac{\partial}{\partial r_{ij}} &= (-1)^{i-1} \delta_{jk} \frac{\partial}{\partial r_k} + \frac{m_i}{M} \delta_{jk} \frac{\partial}{\partial R_k} \\ &= (-1)^{i-1} \frac{\partial}{\partial r_j} + \frac{m_i}{M} \frac{\partial}{\partial R_j}. \end{aligned}$$

So, $\nabla_{\mathbf{r}_i} = (-1)^{i-1} \nabla_{\mathbf{r}} + \frac{m_i}{M} \nabla_{\mathbf{R}}$, where $M = m_1 + m_2$. Thus,

$$\begin{aligned} \nabla_{\mathbf{r}_1}^2 &= \nabla_{\mathbf{r}}^2 + 2 \frac{m_1}{M} \nabla_{\mathbf{r}} \cdot \nabla_{\mathbf{R}} + \left(\frac{m_1}{M} \right)^2 \nabla_{\mathbf{R}}^2 \\ \nabla_{\mathbf{r}_2}^2 &= \nabla_{\mathbf{r}}^2 - 2 \frac{m_2}{M} \nabla_{\mathbf{r}} \cdot \nabla_{\mathbf{R}} + \left(\frac{m_2}{M} \right)^2 \nabla_{\mathbf{R}}^2. \end{aligned}$$

Then, the kinetic energy operator in position space becomes:

$$\begin{aligned} T^{\text{pos}} &= -\frac{\hbar^2}{2m_1} \nabla_{\mathbf{r}_1}^2 - \frac{\hbar^2}{2m_2} \nabla_{\mathbf{r}_2}^2 \\ &= -\frac{\hbar^2}{2} \left\{ \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \nabla_{\mathbf{r}}^2 + \left(\frac{m_1 + m_2}{M^2} \right) \nabla_{\mathbf{R}}^2 \right\} \\ &= -\frac{\hbar^2}{2\mu} \nabla_{\mathbf{r}}^2 - \frac{\hbar^2}{2M} \nabla_{\mathbf{R}}^2. \end{aligned}$$

Or, in operator form:

$$\begin{aligned} \hat{T} &= \hat{T}_{\text{rel}} + \hat{T}_{\text{CM}} \quad \text{with} \\ \hat{T}_{\text{rel}} &= \frac{\hat{\mathbf{p}}^2}{2\mu} \quad \text{and} \quad \hat{T}_{\text{CM}} = \frac{\hat{\mathbf{P}}^2}{2M} \end{aligned}$$

where $\mu = \frac{m_1 m_2}{M}$ is the reduced mass. If the potential only depends on the relative coordinate, then the Hamiltonian takes the form:

$$\begin{aligned} \hat{H} &= \hat{T} + \hat{V} \\ &= \frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} + \hat{V}(\hat{\mathbf{r}}) \\ &= \hat{H}_{\text{rel}} + \hat{H}_{\text{CM}}, \quad \text{where} \end{aligned}$$

$$\begin{aligned}\hat{H}_{\text{rel}} &= \frac{\hat{\mathbf{p}}^2}{2\mu} + \hat{V}(\hat{\mathbf{r}}) \quad \rightarrow \quad H_{\text{rel}}^{\text{pos}} = -\frac{\hbar^2}{2\mu} \nabla_{\mathbf{r}}^2 + V(r) \\ \hat{H}_{\text{CM}} &= \frac{\hat{\mathbf{P}}^2}{2M} \quad \rightarrow \quad H_{\text{CM}}^{\text{pos}} = -\frac{\hbar^2}{2M} \nabla_{\mathbf{R}}^2.\end{aligned}\tag{10.4}$$

This means that the Hamiltonian splits into two independent terms, one acting strictly on the center of mass coordinate and one acting strictly on the relative coordinate. So, the wavefunction factorizes as $\Psi(\mathbf{r}, \mathbf{R}) = \psi(\mathbf{r})\phi(\mathbf{R})$. The center of mass wavefunction includes no potential and thus evolves freely. The relative coordinate wavefunction is the only part that's affected by the potential and is thus the only part that carries the relevant physics.

From the above arguments, one can see that the cancellation of the cross terms between $\mathbf{r}$ and $\mathbf{R}$ is what allows the kinetic energy operator to be factorized, allowing us to separate out the center of mass motion from the relative motion.

10.2.1 Tensor product structure:

To make the tensor product structure explicit, note that it has been implicit from the outset: the two-particle Hilbert space is $\mathcal{H} = L^2(\mathbb{R}_{\mathbf{r}_1}^3) \otimes L^2(\mathbb{R}_{\mathbf{r}_2}^3)$, so states can be written as $|\Psi\rangle = \sum_{i,j} c_{ij} |\psi_i\rangle \otimes |\phi_j\rangle \rightarrow \Psi(\mathbf{r}_1, \mathbf{r}_2) = \sum_{i,j} c_{ij} \psi_i(\mathbf{r}_1) \phi_j(\mathbf{r}_2)$. The Hamiltonian is

$$\hat{H} = \hat{T}_{\mathbf{r}_1} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{T}_{\mathbf{r}_2} + \hat{V}(\hat{\mathbf{r}}_1 \otimes \mathbb{I} - \mathbb{I} \otimes \hat{\mathbf{r}}_2).$$

We introduce a coordinate change on $\mathbb{R}^6$: $(\mathbf{r}_1, \mathbf{r}_2) \rightarrow (\mathbf{r}, \mathbf{R})$. This is a diffeomorphism under which the above calculation shows that $\hat{H}$ transforms to

$$\begin{aligned}\hat{H} &= \hat{T}_{\mathbf{R}} \otimes \mathbb{I} + \mathbb{I} \otimes (\hat{T}_{\mathbf{r}} + \hat{V}_{\mathbf{r}}) \\ &= \hat{H}_{\text{CM}} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{H}_{\text{rel}}.\end{aligned}$$

where $\hat{H}_{\text{CM}}$ acts solely on $L^2(\mathbb{R}_{\mathbf{R}}^3)$ and $\hat{H}_{\text{rel}}$ acts solely on $L^2(\mathbb{R}_{\mathbf{r}}^3)$. Thus, the Hilbert space factorizes as $\mathcal{H} \cong L^2(\mathbb{R}_{\mathbf{r}}^3) \otimes L^2(\mathbb{R}_{\mathbf{R}}^3)$, and $\hat{H}$ respects this factorization.

10.2.2 Extension to n particles:

This extends to n particles via Jacobi coordinates. The kinetic energy $\hat{T} = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_i}$ defines a quadratic form on momentum space with Gram matrix $G_{ij} = \frac{1}{m_i} \delta_{ij}$ (this is just the inverse mass matrix $M = \text{diag}(m_1, \dots, m_n)$). Define a set of linear transformations of the coordinates: $\mathbf{q}_\alpha = \sum_i J_{\alpha i} \mathbf{r}_i$ with $J_{\alpha i} = \frac{\partial \mathbf{q}_\alpha}{\partial \mathbf{r}_i}$ constant. We can do this because we're asserting that we have a linear transformation. If we didn't, the Jacobian would be position-dependent, and the following arguments fall apart. Further, we assume that the Jacobian is non-degenerate, meaning that the coordinate transformation we define produces n new, independent coordinates with no redundancy. Thus, $\mathbf{J}$ is invertible, and $\mathbf{q} = \mathbf{J}\mathbf{r}$ and $\mathbf{r} = \mathbf{J}^{-1}\mathbf{q}$ (where $\mathbf{q}, \mathbf{r} \in \mathbb{R}^{3n}$). The transformation is linear

with constant coefficients, so gradients transform as $\nabla_{\mathbf{r}_\alpha} = \sum_i J_{\alpha i} \nabla_{\mathbf{q}_i}$. Thus,

$$\begin{aligned} \hat{T} = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_i} &\rightarrow T^{\text{pos}} = -\frac{\hbar^2}{2} \sum_i \frac{1}{m_i} \left(\sum_j J_{ij} \nabla_{\mathbf{q}_j} \right)^2 \\ &= -\frac{\hbar^2}{2} \sum_i \frac{1}{m_i} \left(\sum_j J_{ij} \nabla_{\mathbf{q}_j} \cdot \sum_k J_{ik} \nabla_{\mathbf{q}_k} \right) \\ &= -\frac{\hbar^2}{2} \sum_{jk} \left(\sum_i \frac{J_{ij} J_{ik}}{m_i} \right) \nabla_{\mathbf{q}_j} \cdot \nabla_{\mathbf{q}_k} \\ &= -\frac{\hbar^2}{2} \sum_{jk} M_{jk} \nabla_{\mathbf{q}_j} \cdot \nabla_{\mathbf{q}_k} = -\frac{\hbar^2}{2} \nabla_{\mathbf{q}}^T \mathbf{M} \nabla_{\mathbf{q}}, \end{aligned}$$

where $M_{jk} = \sum_i J_{ij} J_{ik} / m_i = (\mathbf{J}^T \mathbf{G} \mathbf{J})_{jk}$ is the reduced mass matrix. If the kinetic energy factorizes in our new coordinate system, $\mathbf{M}$ should be diagonal. This would immediately mean that the cross terms $\nabla_{\mathbf{q}_j} \cdot \nabla_{\mathbf{q}_k}$ don't contribute. We can then choose a $\mathbf{J}$ for which this follows. Here's the procedure: for n particles,

1. Treat particles 1 and 2 as a subsystem, replace them with their CM and relative coordinate. This reduces the n particle system to an $(n-1)$ particle system (the one relative coordinate is now done; we have left the subsystem CM plus the remaining $(n-2)$ particles).
2. Apply the same two-particle argument to the next pair (the subsystem CM and r_3), and so on.
3. Repeat iteratively until all particles are swept over.

The inductive construction works because at each step, the two-particle calculation guarantees diagonalization of that block, and since the steps involve different degrees of freedom, no cross terms are introduced between blocks. Moving through inductively through the entire set of coordinates diagonalizes all blocks, and, by the end, we'll have the full center of mass coordinate of the system, $\mathbf{R}$ and $(n-1)$ relative coordinates $\{\rho_k\}_i^{n-1}$. The induction closes because the two-particle case is already proved. Thus, the kinetic energy can always be made to factorize out the center of mass motion. If the potential only depends on ρ_k , then the full Hamiltonian similarly factorizes.

10.3 Separating out Angular Motion

10.3.1 Overview of Angular Momentum:

Angular momentum $\hat{\mathbf{J}}$ is defined as the generator of rotations. An infinitesimal rotation by angle $\delta\phi$ about axis $\hat{\mathbf{n}}$ acts on states as

$$U(\delta\phi) = 1 - \frac{i}{\hbar} \delta\phi \hat{\mathbf{n}} \cdot \hat{\mathbf{J}},$$

with $\hat{\mathbf{J}}$ Hermitian. This expression so far doesn't restrict the form of $\hat{\mathbf{J}}$; it's just the general form of an infinitesimal transformation operator $U_\epsilon = 1 - i\hat{G}\epsilon$ ($\hat{G}$ Hermitian) that follows from requiring that the transformation be unitary and has certain Lie group properties¹ ($\hat{G} \rightarrow p_x/\hbar, \epsilon \rightarrow dx'$ for infinitesimal translation by dx' in the x direction, for instance). The next step is to restrict the

¹These conditions are as follows:

form of $\hat{G}$ given the specific nature of the transformation. For us, this means using the general properties of rotations to restrict the form of $\hat{\mathbf{J}}$. We require that U obeys the same composition rules as classical rotations (namely, the non-commutativity of rotations about different axes). Demanding this restriction on U , we find $[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k$. This is the Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ (the Lie algebra of $SO(3)$ and its double cover $SU(2)$). Really, this can be thought of as the *definition* of rotations. We arrived at this by demanding that U obeys the same rules as classical rotations, but one can also view infinitesimal rotations as being fundamentally defined by this Lie algebra. Classical rotations don't include spin (classical rotations are strictly defined by $SO(3)$ and not its double cover $SU(2)$), yet quantum mechanics does. Spin is a representation of this same Lie algebra. The Lie algebra was motivated by the behavior of classical rotations, but it includes representations that have no classical analogue.

This Lie algebra defines everything about rotations, regardless of representation or basis. From this structure alone, we can prove that $[\hat{\mathbf{J}}^2, \hat{J}_k] = 0$ for any one k . Thus, $\hat{\mathbf{J}}^2$ and $\hat{J}_k$ are simultaneously diagonalizable with simultaneous eigenkets $|j, m\rangle$ ($[\hat{J}_i, \hat{J}_j] \neq 0$ when $j \neq i$, so we can't simultaneously diagonalize all components of angular momentum at once. We can only add one component to our commuting set). Using ladder operators, we can further show that $m = -j, -j+1, \dots, 0, \dots, j-1, j$; $\hat{\mathbf{J}}^2|j, m\rangle = \hbar^2 j(j+1)|j, m\rangle$; and $\hat{J}_k|j, m\rangle = \hbar m|j, m\rangle$ (i.e., the $\hat{J}_z$ spectrum is discrete with eigenvalues determined by the eigenvalue of $\hat{\mathbf{J}}^2$). Further, this establishes that each j -eigenspace of $\hat{\mathbf{J}}^2$ is $(2j+1)$ -fold degenerate, and $\hat{J}_z$ resolves this degeneracy. Since $[\hat{\mathbf{J}}^2, \hat{J}_k] = 0$ guarantees $\hat{J}_k$ maps each j -subspace to itself, we can diagonalize $\hat{J}_k$ within each block independently, yielding the unique simultaneous eigenbasis $|j, m\rangle$ without mixing different j values. The ladder operator derivation hinted at above is precisely doing this simultaneous diagonalization procedure in disguise.

Spin, $\hat{\mathbf{S}}$, is one example of an angular momentum operator (meaning it satisfies the angular momentum commutation relations). It corresponds to half-integer- j irreps of $\mathfrak{su}(2)$, and generates $SU(2)$ transformations acting on spinor space $\mathbb{C}^2$. This type of rotation has no interpretation as rotations of physical $\mathbb{R}^3$. Orbital angular momentum, $\hat{\mathbf{L}} = \hat{\mathbf{x}} \times \hat{\mathbf{p}}$, is another example. It corresponds to integer- j irreps of $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, and generates $SO(3)$ transformations acting on $\mathbb{R}^3$ (rotations of

-
1. $U(0) = 1$ (identity at zero transformation).
 2. $U(\epsilon_1 + \epsilon_2) = U(\epsilon_1)U(\epsilon_2)$ (group composition). This condition along with condition 1 imply $U(-\epsilon) = U^{-1}(\epsilon)$ (inverse).
 3. Smoothness / differentiability in ϵ near $\epsilon = 0$. This justifies the generator form for finite transformations: $U(\phi) = \exp(-iG\phi)$.
 4. $U^\dagger U = 1$ (unitarity).

The first two are just general group properties. The third is a property that Lie groups must follow, and the final condition, unitarity, is a separate condition all together required to conserve probability. Further, it should be noted that these are not the complete conditions for a Lie group. These are just the sufficient conditions for defining a general infinitesimal transformation operator in quantum mechanics.

physical space)². One can show using the definition $\hat{\mathbf{L}} = \hat{\mathbf{x}} \times \hat{\mathbf{p}}$ that, in position space,

$$\begin{aligned}\langle \mathbf{x} | \hat{\mathbf{L}} | \psi \rangle &= -i\hbar(\mathbf{x} \times \nabla) \langle \mathbf{x} | \psi \rangle \\ \rightarrow \langle \mathbf{x} | \hat{\mathbf{L}}^2 | \psi \rangle &= -\hbar^2 \nabla^2 \langle \mathbf{x} | \psi \rangle.\end{aligned}\tag{10.5}$$

These expressions can be used to find the simultaneous eigenvectors of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$ in the position basis: the spherical harmonics, $\langle \mathbf{x} | l, m \rangle = Y_l^m(\Omega_{\mathbf{x}})$. It's important to note that these eigenfunctions are established completely independently from any Hamiltonian. Their form is completely determined by the structure of $\hat{\mathbf{L}}$, which has nothing to do with $\hat{H}$.

10.3.2 Finding Simultaneous Eigenkets

Now, if $[\hat{\mathbf{L}}, \hat{H}] = 0$, $[\hat{H}, \hat{\mathbf{L}}^2] = 0$ and $[\hat{H}, \hat{L}_k] = 0$, so one can simultaneously diagonalize $\hat{H}$, $\hat{\mathbf{L}}^2$, and $\hat{L}_k$ to find simultaneous eigenkets $|n, l, m\rangle$ where n corresponds to the energy eigenvalue. Note that $[\hat{H}, \hat{\mathbf{L}}^2] = [\hat{H}, \hat{L}_k] = 0$ alone says nothing about how the energy spectrum specified by n relates to l or m (if there exists a relationship at all), if the energy spectrum is discrete, or if the energy spectrum is degenerate. However, $[\hat{\mathbf{L}}, \hat{H}] = 0$, from which we derived $[\hat{\mathbf{L}}^2, \hat{H}] = [\hat{L}_z, \hat{H}] = 0$, does. $[\hat{H}, \hat{\mathbf{L}}] = 0 \rightarrow [\hat{\mathbf{L}}_{\pm}, \hat{H}] = 0$. Thus, if $|n, l, m\rangle$ is an eigenvector of $\hat{H}$ with eigenvalue $E_{n,l,m}$, then

$$\hat{H} \hat{\mathbf{L}}_{\pm} |n, l, m\rangle = \hat{\mathbf{L}}_{\pm} \hat{H} |n, l, m\rangle = E_{n,l,m} \hat{\mathbf{L}}_{\pm} |n, l, m\rangle.$$

So, $\hat{\mathbf{L}}_{\pm} |n, l, m\rangle \propto |n, l, m \pm 1\rangle$ is also an eigenvector with the same energy $E_{n,l,m}$ (an analogous argument can be made using $\hat{\mathbf{L}}_{-}$). So, the energy is completely independent of m , and we should label our energy eigenvalues by $E_{n,l}$. I.e., the energy spectrum includes a $(2l + 1)$ -fold degeneracy that it inherits from $\hat{\mathbf{L}}^2$. Still, we know nothing of if the energy spectrum is discrete or if it's related to l , only that it cannot depend on m .

For standard one-particle Hamiltonians of the form $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(r)$, $[\hat{\mathbf{L}}, \hat{H}] = 0$ if the potential is spherically symmetric, $[\hat{\mathbf{L}}, \hat{V}] = 0$, since the kinetic energy is already spherically symmetric, $[\hat{\mathbf{L}}, \hat{T}] = 0$. For one particle, this amounts to requiring that the potential V is central. For n particles, the argument is as follows: we define the orbital angular momentum of the k -th relative coordinate as $\hat{\mathbf{L}}_{\rho_k} = \hat{\rho}_k \times \hat{\mathbf{p}}_{\rho_k}$ and the orbital angular momentum of the center of mass as $\hat{\mathbf{L}}_{\text{CM}} = \hat{\mathbf{R}} \times \hat{\mathbf{P}}$. Then, we can further define $\hat{\mathbf{L}}_{\text{rel}} = \sum_k \hat{\mathbf{L}}_{\rho_k}$. $[\hat{H}_{\text{CM}}, \hat{\mathbf{L}}_{\text{CM}}] = 0$ trivially (since the center of mass is always free). $[\hat{\mathbf{L}}_{\text{rel}}, \hat{T}_{\text{rel}}] = 0$ for the same reason as the one-particle case: $\hat{T}_{\text{rel}} = \sum_k \hat{T}_{\rho_k}$, and $[\hat{\mathbf{L}}_{\rho_k}, \hat{T}_{\rho_k}] = 0$, so $[\hat{\mathbf{L}}_{\text{rel}}, \hat{T}_{\text{rel}}] = \sum_{i,j} [\hat{\mathbf{L}}_{\rho_i}, \hat{T}_{\rho_j}] = 0$.³ If $\hat{V}(\{\hat{\mathbf{x}}_k\}_k)$ is completely rotationally invariant – i.e., if $V(R\mathbf{x}_1, R\mathbf{x}_2, \dots, R\mathbf{x}_n) = V(\mathbf{x}_1, \dots, \mathbf{x}_n) \forall R \in SO(3)$ (a sufficient but

²There are two ways to arrive at this expression. (1) Take it as a definition. (2) Require that $\hat{\mathbf{L}}$ must generate rotations of the position eigenbasis itself: $|\mathbf{x}\rangle \rightarrow |R\mathbf{x}\rangle$. This is a special restriction because the position eigenkets aren't true states in the Hilbert space. They're generalized states in the Gelfand triple. So, requiring $\hat{\mathbf{L}}$ rotate these states just as it does rotate normal states in the Hilbert space forces $\hat{\mathbf{L}}$ to take a specific form: Taylor expand $\langle \mathbf{x} | U(\delta\phi) | \psi \rangle \approx \psi(\mathbf{x} - \delta\phi \hat{\mathbf{n}} \times \mathbf{x})$ and compare this to the generator expression $U(\delta\phi) | \psi \rangle = (1 - \frac{i}{\hbar} \delta\phi \hat{\mathbf{n}} \cdot \hat{\mathbf{L}}) | \psi \rangle$. This immediately implies $\hat{\mathbf{L}} = \hat{\mathbf{x}} \times \hat{\mathbf{p}}$. One more restriction: since $\mathbf{x}$ are in $\mathbb{R}^3$, a 2π rotation should return every $\mathbf{x}$ to itself. As such, only integer- j irreps of $\mathfrak{su}(2)$ are admitted: the l eigenvalue is restricted to integers (this is easy to see by writing the finite rotation transformation, $\exp(-i\hat{\mathbf{L}} \cdot \hat{\mathbf{n}}\phi/\hbar)$, and noting that this must return to the identity transformation at $\phi = 2\pi$. In the $|l, m\rangle$ basis, this operator acts as $e^{-im\phi}$ (assuming $\hat{\mathbf{n}}$ points along the quantization axis; i.e., the axis which the quantum number m is defined for). Then, for $e^{-i2\pi m} = 1 \forall m, m \in \mathbb{Z}$, and since $m \in \{-l, \dots, l\}$, $l \in \mathbb{Z}$).

³For $i \neq j$, $[\hat{\mathbf{L}}_{\rho_i}, \hat{T}_{\rho_j}] = 0$ because the operators are acting on different Jacobi coordinates. For $i = j$, $[\hat{\mathbf{L}}_{\rho_i}, \hat{T}_{\rho_i}] = 0$ by the same argument as the one-particle case.

not necessary condition for this to hold is that V depends only on pairwise separations $|\mathbf{x}_i - \mathbf{x}_j|$ – then $[\hat{H}_{\text{rel}}, \hat{\mathbf{L}}_{\text{rel}}] = 0$. In this case, $\hat{\mathbf{L}}_{\text{tot}} := \hat{\mathbf{L}}_{\text{CM}} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{\mathbf{L}}_{\text{rel}}$ commutes with $\hat{H}$ because

$$\begin{aligned} [\hat{H}, \hat{\mathbf{L}}_{\text{tot}}] &= [\hat{H}_{\text{CM}} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{H}_{\text{rel}}, \hat{\mathbf{L}}_{\text{CM}} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{\mathbf{L}}_{\text{rel}}] \\ &= [\hat{H}_{\text{CM}}, \hat{\mathbf{L}}_{\text{CM}}] \otimes \mathbb{I} + [\hat{H}_{\text{CM}} \otimes \mathbb{I}, \mathbb{I} \otimes \hat{\mathbf{L}}_{\text{rel}}] + [\mathbb{I} \otimes \hat{H}_{\text{rel}}, \hat{\mathbf{L}}_{\text{CM}} \otimes \mathbb{I}] + \mathbb{I} \otimes [\hat{H}_{\text{rel}}, \hat{\mathbf{L}}_{\text{rel}}] \\ &= 0 + 0 + 0 + 0 = 0. \end{aligned}$$

Lastly, we note that, while we defined $\hat{\mathbf{L}}_{\text{tot}}$ as the sum of the center of mass angular momentum and the relative angular momentum, the more fundamental definition is $\hat{\mathbf{L}}_{\text{tot}} = \sum_i \hat{\mathbf{x}}_i \times \hat{\mathbf{p}}_i$ (the sum over angular momentum for each particle, as in classical mechanics). However, because the Jacobi transformation is a canonical transformation,

$$\hat{\mathbf{L}}_{\text{tot}} = \sum_i \hat{\mathbf{x}}_i \times \hat{\mathbf{p}}_i = \hat{\mathbf{R}} \times \hat{\mathbf{P}} + \sum_k \hat{\rho}_k \times \hat{\mathbf{p}}_{\rho_k}.$$

(the cross terms cancel). Thus, the two definitions are equivalent.

For $n = 2$ particles, the above results are nice because the relative angular momentum is just the angular momentum of the one relative coordinate, and the problem reduces to an effective one-particle problem. Thus, restating the main points: since $[\hat{H}_{\text{rel}}, \hat{\mathbf{L}}_{\text{rel}}^2] = [\hat{H}_{\text{rel}}, \hat{L}_{\text{rel},z}] = [\hat{\mathbf{L}}_{\text{rel}}^2, \hat{L}_{\text{rel},z}] = 0$, $\hat{H}_{\text{rel}}$ is block diagonal in the $|l, m\rangle$ basis (meaning $\hat{H}_{\text{rel}}$ doesn't mix different (l, m) states), and we can label energy eigenkets as simultaneous eigenkets $|n, l, m\rangle$ of $\hat{H}_{\text{rel}}$, $\hat{\mathbf{L}}_{\text{rel}}^2$, and $\hat{L}_{\text{rel},z}$. We now proceed to solve for these eigenkets. First, letting $\mathbf{r}$ be the relative coordinate and $r = |\mathbf{r}|$, recall the relative Hamiltonian from eq. (10.4):

$$\hat{H}_{\text{rel}} = \frac{\hat{\mathbf{p}}^2}{2\mu} + \hat{V}(\hat{\mathbf{r}}) \rightarrow H_{\text{rel}}^{\text{pos}} = -\frac{\hbar^2}{2\mu} \nabla_{\mathbf{r}}^2 + V(r).$$

Since $\nabla_{\mathbf{r}}^2 = \nabla_r^2 + \nabla_{\Omega}^2/r^2$,

$$\begin{aligned} H_{\text{rel}}^{\text{pos}} &= -\frac{\hbar^2}{2\mu} \nabla_{\mathbf{r}}^2 + V(r) \\ &= -\frac{\hbar^2}{2\mu} \nabla_r^2 - \frac{\hbar^2}{2\mu r^2} \nabla_{\Omega}^2 + V(r). \end{aligned}$$

Using eq. (10.5), we can relate this to the angular momentum operator by pulling back to the operator formalism:

$$\begin{aligned} H_{\text{rel}}^{\text{pos}} \psi_{n,l,m}(\mathbf{r}) &= \left(-\frac{\hbar^2}{2\mu} \nabla_r^2 - \frac{\hbar^2}{2\mu r^2} \nabla_{\Omega}^2 + V(r) \right) \langle \mathbf{r} | n, l, m \rangle \\ &= \frac{1}{2\mu} \langle \mathbf{r} | \hat{\mathbf{p}}_r^2 | n, l, m \rangle + \frac{1}{2\mu} \langle \mathbf{r} | \hat{\mathbf{L}}^2 \hat{\mathbf{r}}^{-2} | n, l, m \rangle + \langle \mathbf{r} | \hat{V}(\hat{\mathbf{r}}) | n, l, m \rangle \\ &= \langle \mathbf{r} | \left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hat{\mathbf{L}}^2}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}) \right) | n, l, m \rangle \\ &= \langle \mathbf{r} | \hat{H}_{\text{rel}} | n, l, m \rangle \quad \text{with} \\ \hat{H}_{\text{rel}} &= \frac{\hat{\mathbf{p}}^2}{2\mu} + \hat{V}(\hat{\mathbf{r}}) = \frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hat{\mathbf{L}}^2}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}), \end{aligned}$$

where $\hat{\mathbf{p}}_r \rightarrow \mathbf{p}_r^{\text{pos}} = -i\hbar\nabla_r$ is the radial momentum operator. Now, r and $\Omega_{\mathbf{r}}$ are independent coordinates spanning $\mathbb{R}^3$. As such, the Hilbert space factorizes into a radial and an angular part: $\mathcal{H} = L^2(\mathbb{R}^3) \cong L^2(\mathbb{R}_{>0}, r^2 dr) \otimes L^2(S^2) = \mathcal{H}_r \otimes \mathcal{H}_\Omega$, and every state in $\mathcal{H}$ lives in this tensor product space⁴. Thus, $|n, l, m\rangle$ will be some superposition of product states between states in the radial sector of the Hilbert space and states in the angular sector of the Hilbert space (though, we will see that the structure constrains it further)⁵. From eq. (10.5), we see that $\hat{\mathbf{L}}^2$ acts only on the angular sector, and this is the case for $\hat{L}_z$ as well ($\langle \mathbf{x} | \hat{L}_z | \psi \rangle = -i\hbar \frac{\partial}{\partial \phi} \langle \mathbf{x} | \psi \rangle$). Thus, $\hat{\mathbf{L}}^2$ acts as $\mathbb{I} \otimes \hat{\mathbf{L}}^2$ on the Hilbert space and $\hat{L}_z$ acts as $\mathbb{I} \otimes \hat{L}_z$ on the Hilbert space. The l -eigenspace of $\mathbb{I} \otimes \hat{\mathbf{L}}^2$ is $\mathcal{H}_r \otimes \mathcal{E}_l$ (where $\mathcal{E}_l$ is just the l -eigenspace of $\hat{\mathbf{L}}^2$). The m -eigenspace of $\mathbb{I} \otimes \hat{L}_z$ is $\mathcal{H}_r \otimes \mathcal{E}_m$ (where $\mathcal{E}_m$ is just the m -eigenspace of $\hat{L}_z$). Thus, the joint l, m -eigenspace of $\mathbb{I} \otimes \hat{\mathbf{L}}^2$ and $\mathbb{I} \otimes \hat{L}_z$ is $\mathcal{H}_r \otimes (\mathcal{E}_l \cap \mathcal{E}_m) = \mathcal{H}_r \otimes \text{span}(|l, m\rangle)$. The angular factor is just the $|l, m\rangle$ simultaneous eigenket of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$, as we expect. Thus, $|n, l, m\rangle$ factorizes as a pure product state, $|n, l, m\rangle = |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle$, where $|n, l, m\rangle_{\text{rad}}$ is the part of $|n, l, m\rangle$ that lies in the radial sector, and $|l, m\rangle$, the simultaneous eigenvector of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$, lies completely in the angular sector. Applying this to the TISE⁶:

$$\begin{aligned}
\hat{H}_{\text{rel}}|n, l, m\rangle &= E_{n,l}|n, l, m\rangle \\
\rightarrow \left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{1}{2\mu} \frac{1}{\hat{\mathbf{r}}^2} \otimes \hat{\mathbf{L}}^2 + \hat{V}(\hat{\mathbf{r}}) \otimes \mathbb{I} \right) (|n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle) &= E_{n,l}(|n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle) \\
\rightarrow \frac{1}{2\mu} \hat{\mathbf{p}}_r^2 |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle + \frac{1}{2\mu} \frac{1}{\hat{\mathbf{r}}^2} |n, l, m\rangle_{\text{rad}} \otimes \hat{\mathbf{L}}^2 |l, m\rangle + \hat{V}(\hat{\mathbf{r}}) |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle \\
&= E_{n,l}(|n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle) \\
\rightarrow \frac{\hat{\mathbf{p}}_r^2}{2\mu} |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle + \hat{V}(\hat{\mathbf{r}}) |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle \\
&= E_{n,l} |n, l, m\rangle_{\text{rad}} \otimes |l, m\rangle.
\end{aligned}$$

Projecting onto $\mathbb{I} \otimes \langle l, m |$ (now that we've isolated the (l, m) -block, we project out the $|l, m\rangle$

⁴This isn't trivial; it's the result of the following theorem: for two σ -finite measure spaces (X, μ_X) and (Y, μ_Y) , $L^2(X \times Y, \mu_X \otimes \mu_Y) \cong L^2(X, \mu_X) \otimes L^2(Y, \mu_Y)$. In words, the space of square-integrable functions on a product space is isomorphic to the tensor product of the individual L^2 spaces, meaning any square-integrable function $f(x, y)$ can be written as a countable sum $\sum_i f_i(x)g_i(y)$ converging in the L^2 norm. The key condition is that the measure factorizes as a product measure: $d\mu_{X \times Y} = d\mu_X \otimes d\mu_Y$. For $\mathbb{R}^3$ in spherical coordinates, $\mathbb{R}^3 = \mathbb{R}_{>0} \times S^2$, $d\mu_{\mathbb{R}_{>0}} = r^2 dr$, $d\mu_{S^2} = d\Omega$, and $d\mu_{\mathbb{R}_{>0} \times S^2} = d\mu_{\mathbb{R}_{>0}} \otimes d\mu_{S^2} = r^2 dr d\Omega$. This is a result of functional analysis and measure theory, not quantum mechanics, but we put this here for completeness.

⁵A general state in a tensor product space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ has the form $|\psi\rangle = \sum_{i,j} c_{ij} |a_i\rangle \otimes |b_j\rangle$, where $\{a_i\}_i$ and $\{b_j\}_j$ are bases for the A and B sectors, respectively. In our case, this sum becomes an integral since the basis for the radial and angular sectors of $\mathcal{H}$ are continuous: $|\psi\rangle = \int d\Omega \int r^2 dr \psi(r, \Omega) |r\rangle \otimes |\Omega\rangle$.

⁶One might wonder why the tensor-product structure of the Hilbert space didn't need to be made explicit from the beginning. After all, we initially wrote states as $|n, m, l\rangle$ and operators as $\hat{\mathbf{L}}^2$, rather than as tensor-product objects like $|\psi\rangle_r \otimes |l, m\rangle$ or $\mathbb{I}_r \otimes \hat{\mathbf{L}}^2$. Did we begin with a reduced version of the problem and only later discover additional structure? No. The tensor-product structure was already present implicitly in the fact that the configuration space factorizes into independent coordinates. No new mathematical structure or physics was introduced; we just made explicit a decomposition that was already encoded in the geometry of the problem. In most elementary treatments, this structure remains implicit because the radial and angular degrees of freedom are already built into the coordinate system itself, so writing explicit tensor products adds little computationally. Separation of variables is precisely the process of exploiting this hidden factorization. This should be contrasted with systems involving genuinely distinct degrees of freedom, such as spatial and spin degrees of freedom. There, the tensor-product structure will not remain implicit because operators may couple different sectors of the Hilbert space nontrivially, making expressions like $|\Psi\rangle = |\psi\rangle_{\text{space}} \otimes |\chi\rangle_{\text{spin}}$ and $\hat{A} = \hat{A}_{\text{space}} \otimes \hat{A}_{\text{spin}}$ essential rather than just interpretational.

eigenvector):

$$\begin{aligned} & \frac{\hat{\mathbf{p}}_r^2}{2\mu} |n, l, m\rangle_{\text{rad}} \langle l, m | l, m\rangle + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} |n, l, m\rangle_{\text{rad}} \langle l, m | l, m\rangle + \hat{V}(\hat{\mathbf{r}}) |n, l, m\rangle_{\text{rad}} \langle l, m | l, m\rangle \\ & \qquad \qquad \qquad E_{n,l} |n, l, m\rangle_{\text{rad}} \langle l, m | l, m\rangle \\ & \rightarrow \left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}) \right) |n, l, m\rangle_{\text{rad}} = E_{n,l} |n, l, m\rangle_{\text{rad}}. \end{aligned}$$

Now, it's clear that $|n, l, m\rangle_{\text{rad}}$ must be m -independent given that m shows up nowhere in this projected TISE. Thus, we can relabel $|n, l, m\rangle_{\text{rad}}$ as $|n, l, m\rangle_{\text{rad}} \rightarrow |n, l\rangle$ to make this m -independence explicit. Thus, we've shown that the simultaneous eigenvectors of $\hat{H}_{\text{rel}}$, $\hat{\mathbf{L}}_{\text{rel}}^2$, and $\hat{L}_{\text{rel},z}$ are $|n, l, m\rangle = |n, l\rangle \otimes |l, m\rangle$, where $|l, m\rangle$, the portion of $|n, l, m\rangle$ lying purely in the angular sector, are the simultaneous eigenvectors of $\hat{\mathbf{L}}_{\text{rel}}^2$ and $\hat{L}_{\text{rel},z}$, and $|n, l\rangle$, the portion of $|n, l, m\rangle$ lying purely in the radial sector, is determined strictly by the radial TISE⁷:

$$\left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}) \right) |n, l\rangle = E_{n,l} |n, l\rangle. \quad (10.6)$$

In position space, the radial TISE becomes⁸:

$$\begin{aligned} & \left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}) \right) |n, l\rangle = E_{n,l} |n, l\rangle \\ & \rightarrow \langle r | \left(\frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu \hat{\mathbf{r}}^2} + \hat{V}(\hat{\mathbf{r}}) \right) |n, l\rangle = E_{n,l} \langle r | n, l\rangle \\ & \rightarrow -\frac{\hbar^2}{2\mu} \left(\nabla_r^2 - \frac{l(l+1)}{r^2} - V(r) \right) R_{n,l}(r) = E_{n,l} R_{n,l}(r), \end{aligned}$$

where $R_{n,l} = \langle r | n, l\rangle$ is the radial wavefunction. Note that, since we're in spherical coordinates, $\nabla_r^2 = \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{d}{dr})$. If we make the substitution $R_{n,l}(r) = u_{n,l}(r)/r$, where $u_{n,l}(r)$ is called the reduced radial wave function, we find that the TISE takes the simple form:

⁷The absence of $|l, m\rangle$ the final expression determining $E_{n,l}$ does not mean that the angular sector doesn't contribute to the energy. The absence of $|l, m\rangle$ is just the result of having projected it out. The $|l, m\rangle$ do contribute to the energy through the $\frac{1}{\hat{\mathbf{r}}^2} \otimes \hat{\mathbf{L}}^2$ coupling term in $\hat{H}$, which is why $l(l+1)$ persists even after projection (while the coordinate directions are completely independent, resulting in the factorization of the Hilbert space, the Hamiltonian couples them.). The angular sector isn't energetically inert; it's already been accounted for.

⁸One can also see this using matrix elements, $\langle r | \hat{H}_{r,\text{rel}} | r' \rangle = \left(-\frac{\hbar^2}{2\mu} \nabla_r^2 + V(r) \right) \delta(r - r') = H_{\text{rel}}^{\text{pos},(r)} \delta(r - r')$ and expanding the TISE in the position basis:

$$\begin{aligned} & \hat{H}_{\text{rel}} |n, l\rangle = E_{n,l} |n, l\rangle \\ & \rightarrow \langle r | \int dr' \hat{H}_{r,\text{rel}} | r' \rangle \langle r' | n, l\rangle = E_{n,l} \langle r | n, l\rangle \\ & \rightarrow \int dr' H_{\text{rel}}^{\text{pos},(r)} \delta(r - r') R_{n,l}(r') = E_{n,l} R_{n,l}(r) \\ & \rightarrow H_{\text{rel}}^{\text{pos},r} \int dr' \delta(r - r') R_{n,l}(r') = E_{n,l} R_{n,l}(r) \\ & \rightarrow H_{\text{rel}}^{\text{pos},r} R_{n,l}(r) = E_{n,l} R_{n,l}(r). \end{aligned}$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u_{n,l}}{dr^2} \Big|_r + \left(V(r) + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right) u_{n,l}(r) = E_{n,l} u_{n,l}(r). \quad (10.7)$$

This is called the reduced radial Schödinger equation.

In summary, for a system of one particle with position $\mathbf{r}$ (or two particles with relative position $\mathbf{r}$) and a spherically symmetric potential, $V(\mathbf{r}) = V(r)$, the energy eigenfunctions take the form

$$\begin{aligned} |n, l, m\rangle &= |n, l\rangle \otimes |l, m\rangle \\ \langle \mathbf{r} | n, l, m\rangle &= \langle r | n, l\rangle \otimes \langle \mathbf{e}_r | l, m\rangle \rightarrow \psi_{n,l,m}(\mathbf{r}) = R_{n,l}(r) Y_l^m(\Omega), \end{aligned}$$

where $R_{n,l}(r)$ is determined by the reduced radial SE (eq. (10.7)) and where $Y_l^m(\Omega)$ are the spherical harmonics. The only differences between the 1D and 2D case are:

1. In 1D, m is the mass of the one particle. In 2D, $m \rightarrow \mu = m_1 m_2 / (m_1 + m_2)$ where m_1, m_2 are the masses of the two particles.
2. In 1D, $|n, l, m\rangle$ are simultaneous eigenfunctions of $\hat{H}$, $\hat{\mathbf{L}}^2$, and $\hat{L}_z$. In 2D, $|n, l, m\rangle$ are simultaneous eigenfunctions of $\hat{H}_{\text{rel}}$, $\hat{\mathbf{L}}_{\text{rel}}^2$, and $\hat{L}_{\text{rel},z}$.
3. In 1D, this alone solves the entire system's dynamics. In 2D, this specifically solves for the relative dynamics. But, all of the physics is captured by the relative dynamics, and the full dynamics are captured by appending the CM motion (which we've shown to be free in the cases we care about) to the relative motion: $|\Psi\rangle = |\mathbf{P}\rangle \otimes |n, l, m\rangle$ (simultaneous eigenket of $\hat{H}_{\text{CM}}$ and $\hat{H}_{\text{rel}}$).

Chapter 11

Non-relativistic Scattering Theory Overview

11.1 Setup

In scattering experiments, a beam of particles is shot toward some target, the incoming particles react to the target via some potential, and they leave the target in an outgoing state that's, in general, different from the incoming state. The goal is to relate the asymptotic incoming behavior of the state, $|\psi_{\text{in}}\rangle$, to the asymptotic outgoing behavior of the state, $|\psi_{\text{out}}\rangle$. This idea can represent an incoming particle scattering off some fixed potential, two particles interacting via some shared potential, etc. With Hamiltonian H , the state at $t = 0$, $|\psi\rangle$ evolves as $|\psi\rangle(t) = U(t)|\psi\rangle$ with $U(t) = \exp(-iHt/\hbar)$.

11.2 The S -matrix and T -matrix

We proceed now to find a relationship between $|\psi_{\text{in}}\rangle$, $|\psi_{\text{out}}\rangle$, and $|\psi\rangle$. Consider a Hamiltonian $H = T + V$. For sufficiently well-behaved potentials V , the following theorem holds:

11.2.1 The Asymptotic Condition

Theorem 11.2.1 (Asymptotic Condition:). *For every $|\psi_{\text{in}}\rangle \in \mathcal{H}$, $\exists$ a state $|\psi\rangle \in \mathcal{H}$ s.t.*

$$\lim_{t \rightarrow -\infty} (U(t)|\psi\rangle - U_0(t)|\psi_{\text{in}}\rangle) = 0,$$

where $U_0(t) = \exp(-iH_0t/\hbar)$ is the free-evolution operator. Similarly, $\forall |\psi_{\text{out}}\rangle \in \mathcal{H}$, $\exists |\psi\rangle \in \mathcal{H}$ s.t.

$$\lim_{t \rightarrow \infty} (U(t)|\psi\rangle - U_0(t)|\psi_{\text{out}}\rangle) = 0.$$

In words, for every state $|\psi_{\text{in}}\rangle \in \mathcal{H}$ ($|\psi_{\text{out}}\rangle \in \mathcal{H}$), there is a solution to the Schrödinger equation $U(t)|\psi\rangle$ which approaches the free orbit $U_0(t)|\psi_{\text{in}}\rangle$ ($U_0(t)|\psi_{\text{out}}\rangle$) as $t \rightarrow -\infty$ ($t \rightarrow \infty$). Thus, by multiplying by $U^\dagger$, we see that, given an asymptotic state $|\psi_{\text{in}}\rangle$ ($|\psi_{\text{out}}\rangle$), the state at $t = 0$ that approaches the free orbit $U_0(t)|\psi_{\text{in}}\rangle$ ($U_0(t)|\psi_{\text{out}}\rangle$) is given by

$$|\psi\rangle = \lim_{t \rightarrow -\infty} U^\dagger(t)U_0(t)|\psi_{\text{in}}\rangle = \Omega_+ |\psi_{\text{in}}\rangle \quad |\psi\rangle = \lim_{t \rightarrow \infty} U^\dagger(t)U_0(t)|\psi_{\text{out}}\rangle = \Omega_- |\psi_{\text{out}}\rangle, \quad (11.1)$$

where $\Omega_{\pm} = \lim_{t \rightarrow \mp\infty} U^{\dagger}(t)U_0(t)$ are called the Møller operators. This defines $\Omega_{\pm}$ as isometries on $\mathcal{H}$. Notably, $\Omega_{\pm}^{\dagger}\Omega_{\pm} = I$ but $\Omega_{\pm}\Omega_{\pm}^{\dagger}$ need not equal I . $Im(\Omega_{\pm}) = H_{sc}$ (the scattering subspace of the Hamiltonian). Thus, $P_{\pm} = \Omega_{\pm}\Omega_{\pm}^{\dagger}$ is the orthogonal projector onto this subspace. If $v \in H_{bound}$, then $P_{\pm}v = 0$. If there are no bound states, then $\Omega_{\pm}$ are global isometries, and $\Omega_{\pm}\Omega_{\pm}^{\dagger} = \Omega_{\pm}^{\dagger}\Omega_{\pm} = I$.

11.3 Lippmann-Schwinger Equation

11.4 Partial Wave Expansion

11.5 Analyticity and the Complex Energy Plane

11.6 Wigner Time Delay

11.7 Born Approximation and Validity

Part III

Research Specific Topics

Chapter 12

Discrete Variable Representation

12.1 Introduction

12.2 Matrix Elements

12.3 Quadrature Approximation

12.3.1 Orthogonal Polynomials

12.3.2 Gaussian Quadratures

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Chapter 13

Complex Scaling

Chapter 14

Eigenvector Continuation

Chapter 15

Parametric Matrix Models

Chapter 16

Complex Scaling

Chapter 17

Complex Scaling-Eigenvector Continuation

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